7.2: One-to-One, Onto, and Inverse Functions

Definition.
Let F be a function from a set X to a set Y. F is **one-to-one** (or **injective**) if, and only if, for all elements x_1 and x_2 in X,

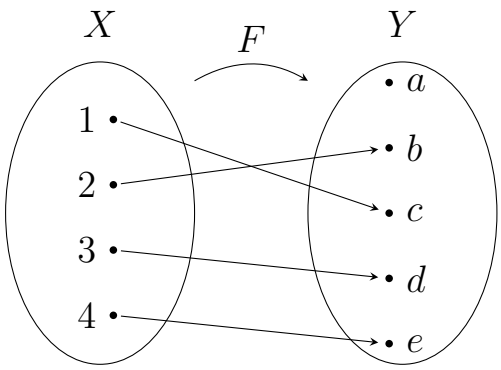
$$\text{if } F(x_1) = F(x_2), \text{ then } x_1 = x_2,$$

or, equivalently,

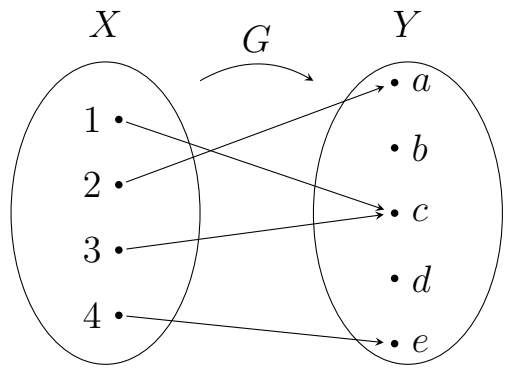
$$\text{if } x_1 \neq x_2, \text{ then } F(x_1) \neq F(x_2).$$

$$F : X \to Y \text{ is one-to-one } \Leftrightarrow \forall x_1, x_2 \in X, \text{ if } F(x_1) = F(x_2) \text{ then } x_1 = x_2.$$

Example. Do either of the arrow diagrams below define a one-to-one function?



One-to-one



NOT one-to-one

since $G(1) = G(3)$

Example. For the following functions, prove they are one-to-one, or show that they are not:

$f(x) = 4x + 2$

suppose $f(x_1) = f(x_2)$

$$4x_1 + 2 = 4x_2 + 2$$
$$-2 \qquad -2$$

$$\frac{4x_1}{4} = \frac{4x_2}{4}$$

$x_1 = x_2 \Rightarrow f$ is injective

$g(s) = \dfrac{s+2}{s-3}$

suppose $g(s_1) = g(s_2)$

$$\frac{s_1+2}{s_1-3} = \frac{s_2+2}{s_2-3}$$

$$\Rightarrow (s_2-3)(s_1+2) = (s_1+2)(s_1-3)$$

$$s_1 s_2 - 3s_1 + 2s_2 - 6 = s_1 s_2 - 3s_2 + 2s_1 - 6$$

$$-3s_1 + 2s_2 = -3s_2 + 2s_1$$

$$-5s_1 = -5s_2$$

$s_1 = s_2 \Rightarrow g$ is injective

$h(t) = \dfrac{e^t + e^{-t}}{2}$

Consider $h(2)$ and $h(-2)$

$$h(2) = \frac{e^2 + e^{-2}}{2} \qquad h(-2) = \frac{e^{-2} + e^2}{2}$$

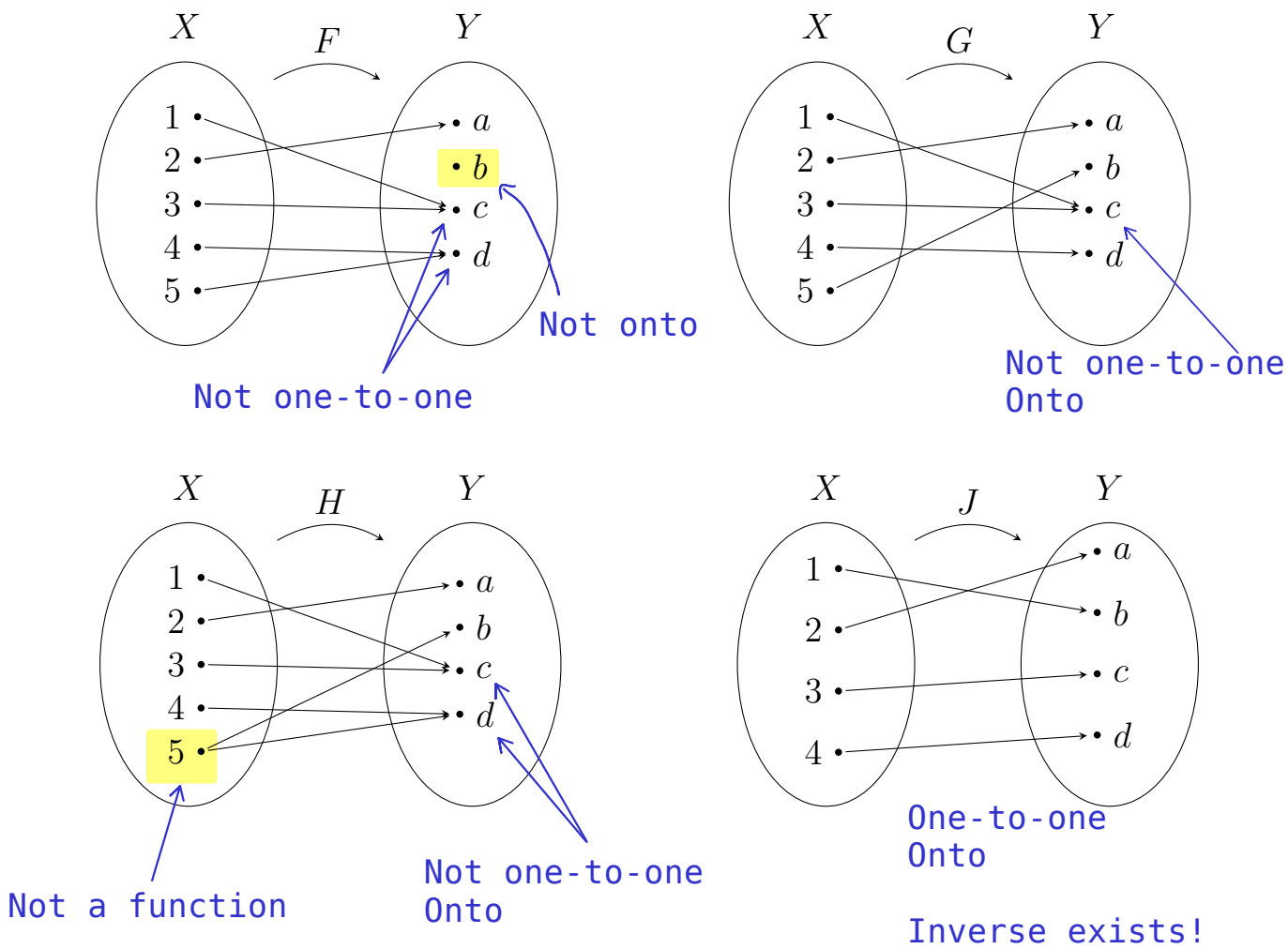
so $h(2) = h(-2)$, but $2 \neq -2$

$\Rightarrow$ NOT injective

Definition.
Let F be a function from a set X to a set Y. F is **onto** (or **surjective**) if, and only if, given any element y in Y, it is possible to find an element x in X with the property that $y = F(x)$

$$F : X \to Y \text{ is onto } \Leftrightarrow \forall y \in Y,\ \exists x \in X \text{ such that } F(x) = y.$$

Example. Do any of the arrow diagrams below define a function that is onto?



Example. For the following functions, prove they are onto, or show that they are not:

$f : \mathbb{R} \to \mathbb{R}$ where $f(x) = 4x - 1,\ \forall x \in \mathbb{R}$

$y = 4x - 1$

$y + 1 = 4x$

$\frac{y+1}{4} = x$

Given any Real number y,

$x = \frac{y+1}{4}$ is a Real number.

Thus f is onto.

(It's possible to pick $x \in \mathbb{R}$ such that $f(x) = y, \forall y \in \mathbb{R}$)

$g : \mathbb{Z} \to \mathbb{Z}$ where $g(n) = 4n - 1,\ \forall n \in \mathbb{Z}$

$m = 4n - 1$

$m + 1 = 4n$

$\frac{m+1}{4} = n$

Consider the integer $m = 1$.

Then $n = \frac{1+1}{4} = \frac{2}{4} = \frac{1}{2} \notin \mathbb{Z}$

$\Rightarrow$ There exists $m \in \mathbb{Z}$ such that $g(n) \neq m \ \forall n \in \mathbb{Z}$

Definition.
A **one-to-one correspondence** (or **bijection**) from a set X to a set Y is a function $F : X \to Y$ that is *both* one-to-one and onto.

If F is a one-to-one correspondence, then F^{-1} exists:

$$F(x) = y \Leftrightarrow F^{-1}(y) = x$$

Example. Find the inverse of the following functions:

$f(x) = 4x + 2$

Swap x & y

$y = 4x + 2 \longrightarrow x = 4y + 2$

Solve for y

$x - 2 = 4y$

$\boxed{\frac{x-2}{4} = y}$

$g(s) = \dfrac{s+2}{s-3}$

Swap s & t

$t = \frac{s+2}{s-3} \rightarrow s = \frac{t+2}{t-3}$

Solve for t

$s(t-3) = t+2$

$st - 3s = t + 2$

$st - t = 3s + 2$

$t(s-1) = 3s + 2$

$\boxed{t = \frac{3s+2}{s-1}}$